Performance of Local Open-Weight Large Language Models for Adjudicating Atrial Fibrillation Hospitalizations

Methods

Study design and data source

We performed a retrospective evaluation using deidentified records from **MIMIC-IV version 3.1**, which contains hospital and critical-care data from Beth Israel Deaconess Medical Center [3,4]. Cohort extraction and preprocessing were performed with Google BigQuery, and local model inference was conducted in Google Colab. One hospitalization per patient was retained to prevent repeat patients from contributing correlated observations.

Cohort construction

The source population contained 546,028 adult admissions among 223,452 patients. Restricting to ICU or emergency/observation care yielded 460,447 admissions among 203,976 patients. Requiring a complete discharge summary and retaining one admission per patient yielded 124,383 eligible admissions and patients. The final evaluation sample included 250 unique patients and hospitalizations: 75 with an any-position atrial fibrillation (AF) ICD code and 175 without one. The exploratory principal-AF ICD proxy was positive in 4 hospitalizations and negative in 246.

Table 1. Cohort construction

Stage	Admissions	Patients
Adult admissions (age ≥18 years)	546,028	223,452
Adult ICU or ED admissions	460,447	203,976
Complete full-text summary; one admission retained per patient	124,383	124,383

Stage	Admissions	Patients
Final analytic sample	250	250

Eligibility criteria

Eligible hospitalizations were adult ICU or emergency/observation encounters with a complete discharge summary of at least 1,500 characters containing HPI and hospital-course sections. AF ICD codes were used to enrich sampling, not to exclude all records without an AF code. Isolated atrial flutter was excluded unless atrial fibrillation was also present. The Fusion condition additionally used linked AF-management medication and procedure context.

Table 2. Eligibility criteria

Domain	Inclusion	Exclusion
Age	Age ≥18 years	Age <18 years
Care setting	ICU or ED/emergency/observation hospitalization	No qualifying ICU or ED encounter
Discharge summary	Full text; ≥1,500 characters; HPI and hospital-course sections present	Missing, short, partial, stub, truncated, or missing required sections
AF evidence source	Discharge-summary narrative; linked medication and procedure context for Fusion	No AF documentation required for the non-enriched comparison stratum
AF ICD enrichment	ICD-9-CM 42731 or ICD-10-CM I480/I481*/I482*/I4891 used to oversample probable AF; isolated atrial flutter excluded unless AF was also present	Absence of an AF code did not exclude a case from the non-enriched stratum
Deduplication	One hospitalization retained per patient by deterministic selection	Repeat admissions excluded after selection

Model selection and local inference

We evaluated four instruction-tuned, open-weight models from the Hugging Face Hub: google/gemma-3-4b-it, google/gemma-3-12b-it, Qwen/Qwen3-8B, and Qwen/Qwen3-32B. Gemma models used full bfloat16 precision; Qwen models used 4-bit NF4 quantization with double quantization, bfloat16 compute, and thinking disabled. All inference ran locally in Google Colab on an NVIDIA A100 40 GB GPU. Every model evaluated the same 250 hospitalizations under all four prompts (1,000 predictions per model; 4,000 total).

Table 3. Technology and reproducibility specification

Component	Specification
Dataset	MIMIC-IV version 3.1; deidentified ICU and hospital EHR data
Cohort extraction	Google BigQuery (credentialed access); https://physionet.org/content/mimiciv/3.1/
Notebook environment	Google Colab
Accelerator	NVIDIA A100 40 GB GPU; full bfloat16 for Gemma 3 and 4-bit NF4 for Qwen3
Models	Hugging Face Hub: google/gemma-3-4b-it; google/gemma-3-12b-it; Qwen/Qwen3-8B; Qwen/Qwen3-32B
Inference location	User-controlled Google Colab runtime; no external hosted-model API used
Evidence and output	Discharge summary for all prompts; linked AF medication/procedure context additionally for Fusion; exactly YES or NO

Prompt strategies

Each model received a system instruction requiring exactly one output word, YES or NO. Four user-prompt strategies were tested: Direct, Role, Structured, and Fusion. All conditions used the discharge summary; Fusion additionally appended linked AF-management medication-order and procedure context. These linked signals were explicitly treated as nonspecific and interpreted only with the narrative. The exact instructions are reproduced in Table 4.

Table 4. Exact prompts

Component	Exact instruction or construction
System prompt	Use only the supplied deidentified MIMIC-IV record. Do not reveal reasoning. Return only YES or NO and no explanation.
Direct	Did this hospitalization involve atrial fibrillation as a documented active clinical problem? Do not count a remote history alone. Answer with a single word: YES or NO.
Role	You are a board-certified cardiologist and clinical-trial endpoint adjudicator. Determine whether atrial fibrillation was an active clinical problem during this hospitalization, rather than only a remote or incidental historical diagnosis. Answer with a single word: YES or NO.
Structured	Adjudicate whether atrial fibrillation was an active problem during this hospitalization. Classify as YES when the discharge summary supports one or more of the following: 1. New, recurrent, worsening, or symptomatic atrial fibrillation contributed to presentation or admission; OR 2. Atrial fibrillation

Component	Exact instruction or construction
	required active inpatient rate control, rhythm control, cardioversion, ablation, anticoagulation decision-making, or other hospital-level management; OR 3. Atrial fibrillation materially affected the hospital course. Do not classify as YES for a remote history or stable problem-list mention alone. Do not infer AF from age, stroke, heart failure, anticoagulation, or tachycardia without documentation linking it to AF. Answer with a single word: YES or NO.
Fusion	Adjudicate whether atrial fibrillation was an active problem during this hospitalization. Use BOTH the discharge summary and the linked AF-management context supplied below. The linked context may include rate/rhythm-control drugs, anticoagulants, cardioversion, ablation, or electrophysiology procedures, but these are nonspecific and must be interpreted with the narrative. Do not count a remote history alone and do not infer AF from a medication by itself. Answer with a single word: YES or NO.
Input construction	The selected prompt was followed by the discharge-summary evidence wrapper. Linked medication-order and procedure context was appended only for Fusion.

Outcomes and statistical analysis

The primary outcome was accuracy against an any-position AF ICD administrative proxy. Accuracy is reported with Wilson 95% confidence intervals. Secondary outcomes were sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), Cohen's κ, and confusion counts. A principal-AF ICD proxy was analyzed exploratorily but was not used for model ranking because only 4 of 250 cases were positive. Operational outcomes included requests, scored outputs, invalid responses, failed calls, and mean and median per-case inference latency.

Ethics and data governance

MIMIC-IV is a deidentified credentialed-access database. This project used local inference so that hospitalization text was not submitted to an external model API. Access to MIMIC-IV remains subject to PhysioNet credentialing, data-use requirements, and institutional policy.

Results

Cohort

The analytic cohort comprised 250 unique adult hospitalizations: 75 positive and 175 negative by the any-position AF ICD proxy, for a proxy-positive prevalence of 30.0%. The exploratory principal-AF proxy identified only 4 positive and 246 negative hospitalizations (1.6%).

Model and prompt performance

Across 16 model-prompt combinations, accuracy ranged from 76.8% to 96.0% (Table 1 and Figure 1). Gemma 3 12B with Direct was best overall at 96.0% (95% CI, 92.8%-97.8%), with 89.3% sensitivity, 98.9% specificity, and κ=0.903. Qwen3 32B Direct reached 95.6% accuracy (92.3%-97.5%); Qwen3 8B Fusion reached 95.2% (91.8%-97.2%) with 99.4% specificity and only one false positive; and Gemma 3 4B Fusion reached 89.6% (85.2%-92.8%). Figure 2 compares the best prompt for each model.

Table 1. Agreement with the any-position AF ICD administrative proxy

Model	Prompt	Accuracy (95% CI)	Sensitivity	Specificity	PPV	NPV	Cohen's κ
Gemma 3 4B	Direct	87.6% (82.9–91.1)	66.7%	96.6%	89.3%	87.1%	0.682
Gemma 3 4B	Role	78.0% (72.5–82.7)	82.7%	76.0%	59.6%	91.1%	0.528
Gemma 3 4B	Structured	76.8% (71.2–81.6)	78.7%	76.0%	58.4%	89.3%	0.497
Gemma 3 4B	Fusion	89.6% (85.2–92.8)	73.3%	96.6%	90.2%	89.4%	0.738
Gemma 3 12B	Direct	96.0% (92.8–97.8)	89.3%	98.9%	97.1%	95.6%	0.903
Gemma 3 12B	Role	90.0% (85.7–93.1)	86.7%	91.4%	81.3%	94.1%	0.766
Gemma 3 12B	Structured	94.8% (91.3–96.9)	89.3%	97.1%	93.1%	95.5%	0.875
Gemma 3 12B	Fusion	93.6% (89.9–96.0)	85.3%	97.1%	92.8%	93.9%	0.844
Qwen3 8B	Direct	93.6% (89.9–96.0)	88.0%	96.0%	90.4%	94.9%	0.846
Qwen3 8B	Role	81.2% (75.9–85.6)	88.0%	78.3%	63.5%	93.8%	0.597
Qwen3 8B	Structured	94.0% (90.3–96.3)	85.3%	97.7%	94.1%	94.0%	0.853
Qwen3 8B	Fusion	95.2% (91.8–97.2)	85.3%	99.4%	98.5%	94.1%	0.881
Qwen3 32B	Direct	95.6% (92.3–97.5)	89.3%	98.3%	95.7%	95.6%	0.893
Qwen3 32B	Role	94.0% (90.3–96.3)	86.7%	97.1%	92.9%	94.4%	0.854
Qwen3 32B	Structured	90.8% (86.6–93.8)	73.3%	98.3%	94.8%	89.6%	0.766
Qwen3 32B	Fusion	94.8% (91.3–96.9)	90.7%	96.6%	91.9%	96.0%	0.876

Prompt effects were model dependent. Fusion was the highest-accuracy strategy for Gemma 3 4B and Qwen3 8B, improving on Direct by 2.0 and 1.6 percentage points, respectively. Direct was best for Gemma 3 12B and Qwen3 32B. Thus, Fusion performed very well in selected models rather than dominating globally. Under the exploratory principal-AF proxy, all configurations detected all four positives, but specificity ranged from 59.3% to 78.9%, accuracy from 60.0% to 79.2%, and κ from 0.045 to 0.107; the sparse positive count precluded meaningful ranking.

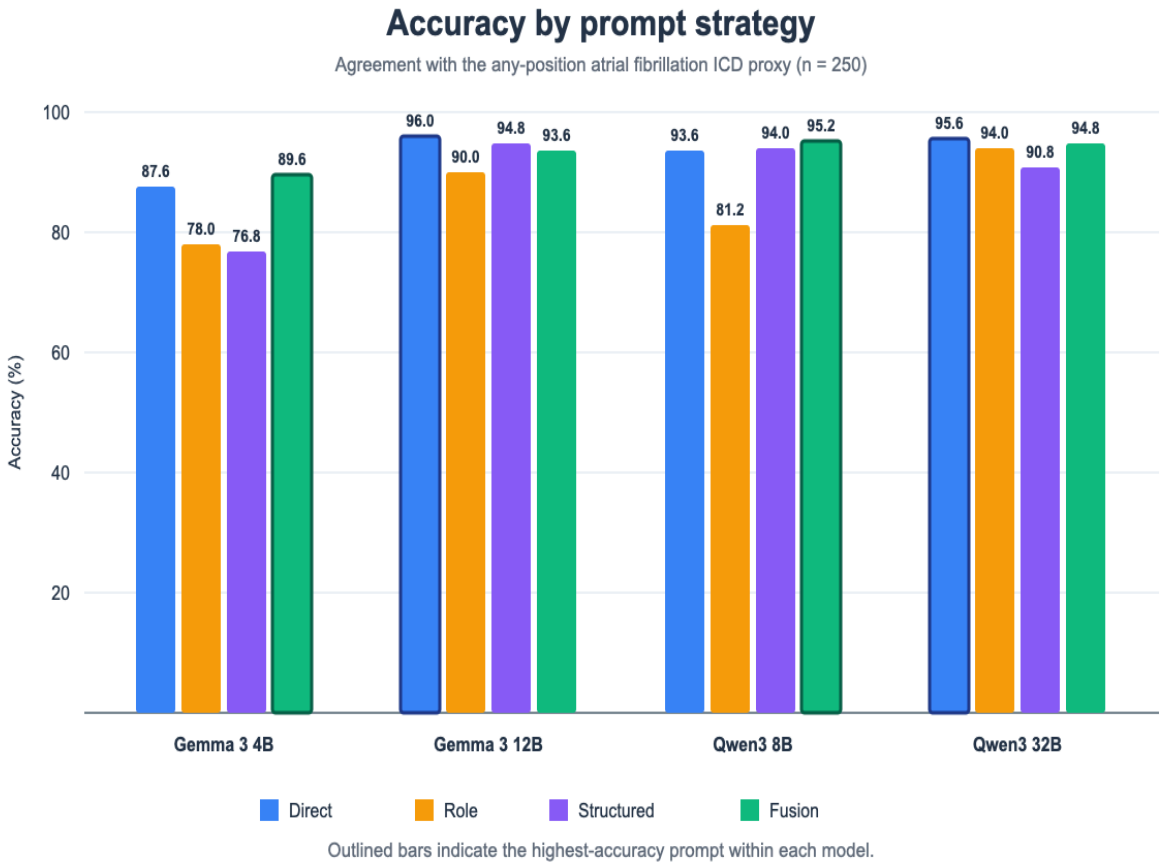


Figure 1. Accuracy by prompt strategy against the any-position AF ICD proxy. Grouped comparison of Direct, Role, Structured, and Fusion for all four models.

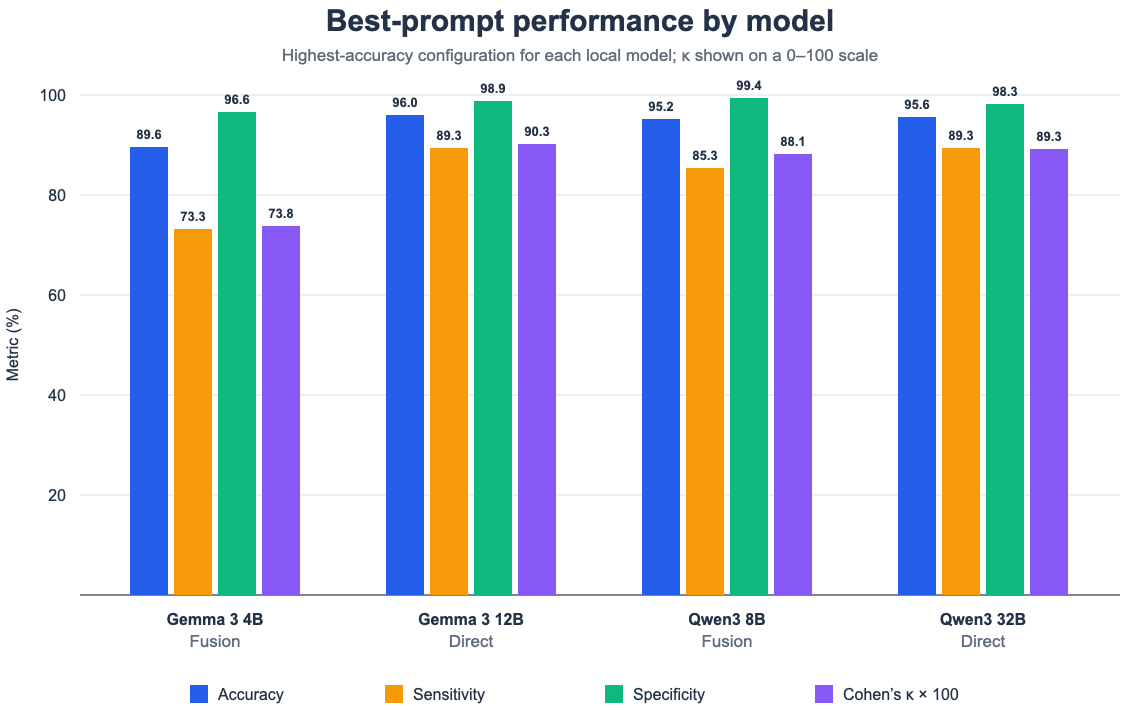


Figure 2. Best-prompt AF performance by model. Accuracy, sensitivity, specificity, and Cohen's κ for the highest-accuracy prompt within each model.

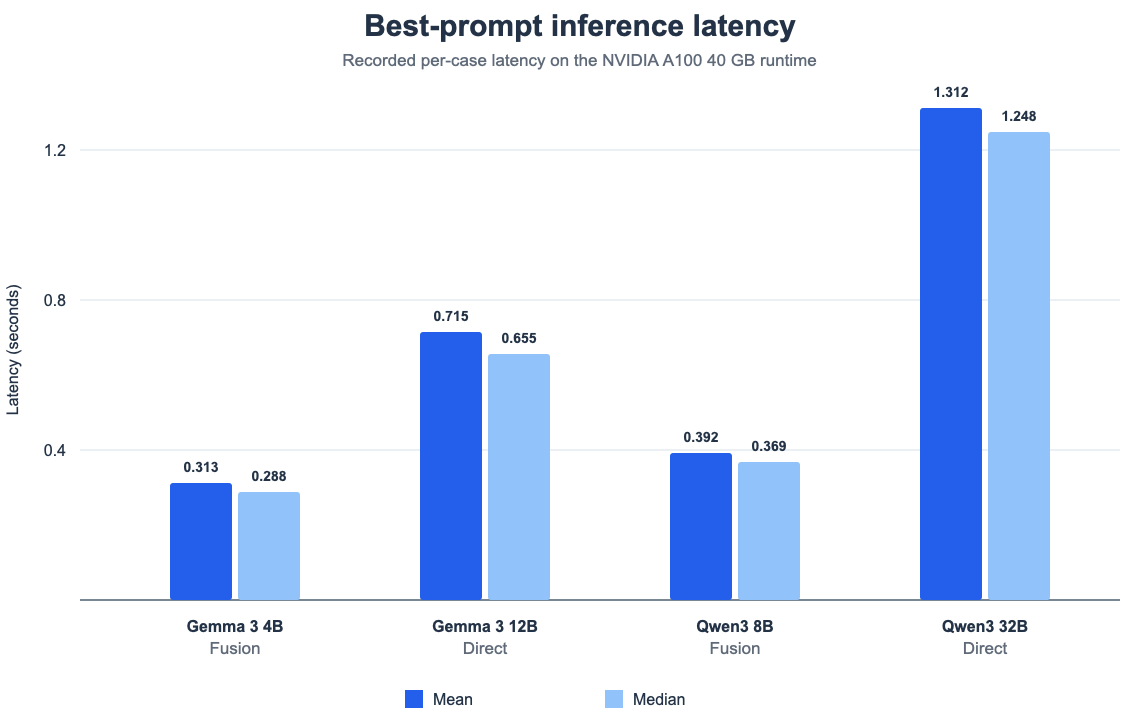


Figure 3. Best-prompt AF inference latency. Mean and median recorded per-case latency for the highest-accuracy prompt within each model.

Confusion counts and operational performance

All 4,000 classifications were scored, with zero invalid responses, zero failed calls, and zero Qwen thinking blocks. Mean latency across configurations ranged from 0.306 to 1.356 seconds per case (Table 2). Among each model's best prompts, Qwen3 8B Fusion achieved 95.2% accuracy and κ=0.881 at a mean latency of 0.392 seconds—0.8 percentage points below Gemma 3 12B Direct while approximately 1.8 times faster. Gemma 3 4B Fusion was fastest at 0.313 seconds but was 6.4 percentage points less accurate than the overall best. Figure 3 summarizes the tradeoff.

Table 2. Confusion counts and recorded per-case inference latency

Model	Prompt	TP	TN	FP	FN	Mean latency, s	Median latency, s
Gemma 3 4B	Direct	50	169	6	25	0.309	0.283
Gemma 3 4B	Role	62	133	42	13	0.306	0.284
Gemma 3 4B	Structured	59	133	42	16	0.314	0.290
Gemma 3 4B	Fusion	55	169	6	20	0.313	0.288
Gemma 3 12B	Direct	67	173	2	8	0.715	0.655

Model	Prompt	TP	TN	FP	FN	Mean latency, s	Median latency, s
Gemma 3 12B	Role	65	160	15	10	0.717	0.661
Gemma 3 12B	Structured	67	170	5	8	0.740	0.680
Gemma 3 12B	Fusion	64	170	5	11	0.738	0.675
Qwen3 8B	Direct	66	168	7	9	0.381	0.362
Qwen3 8B	Role	66	137	38	9	0.383	0.364
Qwen3 8B	Structured	64	171	4	11	0.393	0.368
Qwen3 8B	Fusion	64	174	1	11	0.392	0.369
Qwen3 32B	Direct	67	172	3	8	1.312	1.248
Qwen3 32B	Role	65	170	5	10	1.318	1.257
Qwen3 32B	Structured	55	172	3	20	1.356	1.277
Qwen3 32B	Fusion	68	169	6	7	1.351	1.275